

SmartBasket AI Technical Product Blueprint v4 – Addendum

This addendum is intended to be appended to the existing v3 blueprint and expands it with production-grade architecture and implementation guidance.

Website-First Strategy

The SmartBasket website is the primary product. All engineering effort should prioritise the responsive Next.js web application. Mobile support should initially be delivered as a Progressive Web App (PWA). A native iOS/Android application using React Native + Expo is a future phase that reuses the same backend, database and AI services.

Production Architecture

Frontend: Next.js + React + TypeScript + Tailwind CSS. Backend: Next.js API Routes. Database: Supabase PostgreSQL. Authentication: Supabase Auth. Hosting: Vercel. AI: Claude/OpenAI APIs. Images stored in Supabase Storage. GitHub is the source of truth with automatic deployment to Vercel.

Retail Data Pipeline

Product information should not be collected live during user requests. Instead, build a dedicated Retail Data Pipeline that periodically collects product information, validates it, normalises it and stores it in Supabase. The website reads only from Supabase. The pipeline can initially use automated collection where appropriate and later migrate to official retailer APIs or licensed feeds without changing the website.

Database Design

Separate Products from Store Listings. Products contain nutrition, allergens, ingredients, barcode and images. Store Listings contain retailer, price, unit price, availability, promotions and price history. Additional tables include Users, Pantry, Shopping Lists, Meal Plans, Recipes, Recipe Ingredients, AI Recommendations and Saved Baskets.

Professional Recipe System

Maintain a structured recipe catalogue with categories including breakfast, lunch, dinner, snacks, desserts, drinks, meal prep, vegetarian, vegan, gluten-free, dairy-free, budget meals and high-protein meals. Each recipe stores ingredients, quantities, preparation steps, cooking time, nutrition, allergens, servings, equipment and substitution suggestions.

Recommended Data Sources

Use a hybrid approach: Open Food Facts for packaged food metadata where appropriate, USDA FoodData Central for nutrient reference data, and a curated first-party SmartBasket recipe collection. AI should personalise recipes and meal plans rather than invent nutritional facts.

Deployment Roadmap

Phase 1: Develop and deploy the website. Phase 2: Convert to PWA. Phase 3: Native mobile application. Implement CI/CD with GitHub and Vercel, HTTPS, environment variables, monitoring, backups and logging.